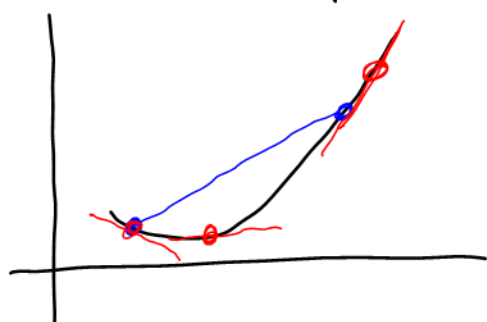


Lecture 18: Concavity and 2nd Derivative Test

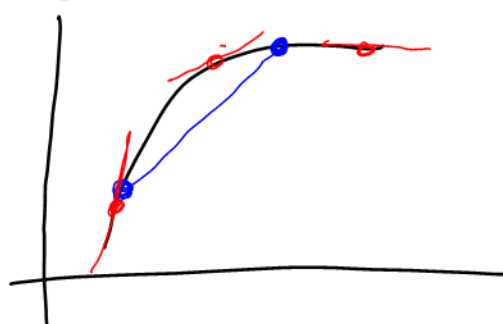
Goal: Use the 2nd derivative to determine properties of the original function.

Concave Up [CU]



f' is increasing
 $[f']' = f'' > 0$

Concave Down [CD]



f' is decreasing
 $[f']' = f'' < 0$

Theorem (C Test) Let f be a fcn where f'' exists on an open interval I

(1) If $f''(x) > 0$ for all x in I , f is CU on I .

(2) If $f''(x) < 0$ for all x in I , f is CD on I .



Example 1 Discuss the shape of $f(x) = x^3 - 12x + 1$



Step 1 Find $f' = 0$ and $f'' = 0$
 $f'(x) = 3x^2 - 12 \stackrel{\text{set } 0}{=} 3(x-2)(x+2) \stackrel{\text{set } 0}{=}$

$\Rightarrow x = \pm 2$

$f''(x) = 6x \stackrel{\text{set } 0}{=} 0 \Rightarrow x = 0$

Step 2 Make sign chart for f' and f''

		-2		0		2		100
Test Point	-100		-1		1		2	
Sign of f' $3(x-2)(x+2)$	+		-		-		+	
Sign of $6x = f''$	-		-		+		+	
Results of IP Test	Inc		Dec		Dec		Inc	
Results of C Test	CD		CD		CU		CU	



Conclusion: f is increasing and CD on $(-\infty, -2)$
 f is decreasing and CD on $(-2, 0)$
 f is decreasing and CU on $(0, 2)$
 f is increasing and CU on $(2, \infty)$



Def If f changes concavity at a point P , then P is called an inflection point.

Theorem If $(c, f(c))$ is an inflection point, either $f''(c) = 0$ or undefined

Why? IPs are critical numbers of f'



If the reverse is not true.

Example 2 Determine the location of the inflection points for
 $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1$

Step 1 Solve $f''(x) = 0$

$$f'(x) = 12x^3 - 12x^2 - 12x + 12$$

$$f''(x) = 12 \cdot 3x^2 - 12 \cdot 2x - 12 \cdot 1 \stackrel{\text{set}}{=} 0$$

$$12(3x^2 - 2x - 1) = 0$$

$$12(x-1)(3x+1) = 0$$

$$\Rightarrow x = -\frac{1}{3}, 1$$

Step 2 Make Sign Chart for f''

		$-\frac{1}{3}$	0	1	100
Test Point	-100				
Sign of 12	+		+		+
Sign of $x-1$	-		-		+
Sign of $3x+1$	-		+		+
Sign of f''	+		-		+
Result of C-Test	CU		CP		CU

$\Rightarrow$ IPs located at $x = -\frac{1}{3}, 1$

Example 3 Find all IPs of

$$f(x) = \frac{1}{20}x^5 - \frac{1}{6}x^4 + \frac{1}{6}x^3 - x + 42$$

Step 1 $f''(x) = 0$ or undefined

$$f'(x) = \frac{1}{4}x^4 - \frac{2}{3}x^3 + \frac{1}{2}x^2 - 1$$

$$\begin{aligned} f''(x) &= x^3 - 2x^2 + x \\ &= x(x^2 - 2x + 1) \\ &= x(x-1)^2 \quad \underline{\text{set } 0} \end{aligned}$$

$$\Rightarrow x = 0, 1$$

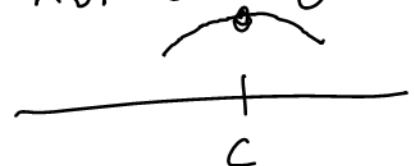
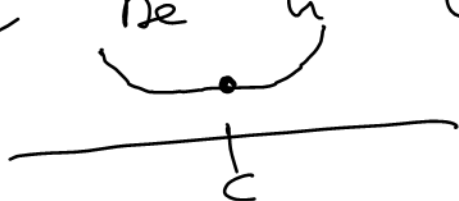
Step 2 Make Sign Chart for f''

	$\leftarrow \begin{array}{c} & & \\ 0 & & 1 \end{array} \rightarrow$		
Test Point	-100	$\frac{1}{2}$	100
Sign of x	-	+	+
Sign of $(x-1)^2$	+	+	+
Sign of f''	-	+	+
C Test	CD	CU	CU

I.e., there is only one IP at $(0, f(0)) = (0, 42)$

Concavity can be used to detect rel. ext.

Let c be a critical number of f



$$f''(c) > 0 \Rightarrow f \text{ is CU near } c \quad | \quad f''(c) < 0 \Rightarrow f \text{ is CP}$$

Theorem (2nd Derivative Test) Let f'' be

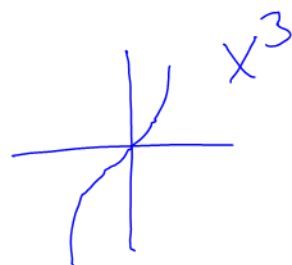
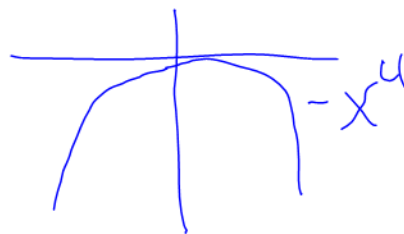
continuous near c and $f'(c) = 0$

(1) If $f''(c) > 0$, f has a rel. min. at c

(2) If $f''(c) < 0$, f has a rel. max. at c

(3) If $f''(c) = 0$, the test is inconclusive
[e, pick another test]

Why (3)?



Example 2 Locate rel. ext. for
 $f(x) = x^3 - 3x^2$

Step 1 Find crit. nums.

$$f'(x) = 3x^2 - 6x = 3x(x-2) \stackrel{\text{set}}{=} 0$$

$$\Rightarrow x = 0, 2$$

Step 2 Determine the sign of $f''(0)$ and $f''(2)$

$$f''(x) = 6x - 6 = 6(x-1)$$

$$f''(0) = -6 < 0 \quad \text{---} \Rightarrow f \text{ has a rel. max at } 0$$

$$f''(2) = 6 > 0 \quad \text{++} \Rightarrow f \text{ has a rel. min at } 2$$

Ex 5 $f(x) = e^x (x^2 - 5x - 13)$

$$f'(x) = e^x (x - 6)(x + 3) \stackrel{\text{set}}{=} 0$$

$$\Rightarrow -3, 6$$

$$f''(x) = e^x (x^2 - x - 21)$$

$$\frac{x = -3:}{\uparrow \text{max } f} f''(-3) = e^{-3} (9 + 3 - 21) < 0 \quad \cap$$

$$x = 6: \underset{\uparrow \text{min}}{f''(6)} = e^6 (36 - 6 - 21) > 0 \quad \cup$$